ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

# INTRODUCTION TO TECHNICAL PROGRAMMING

## PAPER 2 · PRACTICE D · MARKING GUIDELINE

**MARKS: 100**

**Marking guideline for the Academic Central NC(V) Level 3 Introduction to Technical Programming paper 2 · practice d.**

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training marking guideline.

**How to mark:** each ✓ is one mark. Programs are not marked strictly on syntax: give marks for correct logic, correct variables and correct output. Accept other variable names, other prompts and other correct ways of solving the problem.

## QUESTION 1

**1.1** Project created and named correctly ✓ (1)

**1.2** Suitable sprite ✓ (1)

**1.3** Suitable backdrop ✓ (1)

**1.4** Variables created ✓✓✓ (percent, level, aps) and set to 0 ✓✓✓ (6)

**1.5** repeat 6 ✓ around the subject blocks ✓ (2)

**1.6** ask block ✓; set percent to answer ✓ (2)

**1.7** Correct conditions for levels 7 ✓, 6 ✓, 5 ✓, 4 ✓, 3 ✓ and 2 ✓; level 1 in the last else ✓; correct nesting of if … else ✓✓; boundaries correct (e.g. 80 gives 7, 79 gives 6) ✓ (10)

**1.8** change aps by level ✓✓ (2)

**1.9** say the level ✓ joined with text ✓ (2)

**1.10** say the total APS after the loop ✓✓ (2)

**1.11** Levels 7, 6, 5, 4, 3, 2 ✓✓; APS = 27 ✓ (3)

```
when green flag clicked
set [percent] to 0
set [level] to 0
set [aps] to 0
repeat 6
    ask [Enter the percentage for a subject] and wait
    set [percent] to (answer)
    if < (percent) > 79 > then
        set [level] to 7
    else
        if < (percent) > 69 > then
            set [level] to 6
        else
            if < (percent) > 59 > then
                set [level] to 5
            else
                if < (percent) > 49 > then
                    set [level] to 4
                else
                    if < (percent) > 39 > then
                        set [level] to 3
                    else
                        if < (percent) > 29 > then
                            set [level] to 2
                        else
                            set [level] to 1
```

```
    say (join [Level: ] (level)) for 2 seconds
    change [aps] by (level)
end repeat
say (join [Your APS is ] (aps)) for 3 seconds
```

**[32]**

## QUESTION 2

**2.1** File created and named correctly ✓ (1)

**2.2** #include <iostream> ✓; using namespace std; ✓ (2)

**2.3** int main() ✓; units, amount, vat, total declared as double ✓✓ and set to 0 ✓ (4)

**2.4** Prompt and cin ✓✓; if units < 0 ✓ show "Invalid units" and return / end ✓ (4)

**2.5** if (units <= 350) ✓✓ amount = units * 2.50 ✓✓; else ✓ amount = 350 * 2.50 + (units − 350) * 3.10 ✓✓✓ (8)

**2.6** vat = amount * 0.15 ✓✓; total = amount + vat ✓ (3)

**2.7** Amount ✓, VAT ✓ and total ✓ displayed with R (3)

```
// YourName - Question 2
#include <iostream>
using namespace std;

int main() {
    double units = 0, amount = 0, vat = 0, total = 0;

    cout << "Enter the units used (kWh): ";
    cin >> units;
    if (units < 0) {
        cout << "Invalid units" << endl;
        return 0;
    }

    if (units <= 350) {
        amount = units * 2.50;
    } else {
        amount = 350 * 2.50 + (units - 350) * 3.10;
    }
    vat = amount * 0.15;
    total = amount + vat;

    cout << "Amount before VAT: R" << amount << endl;
    cout << "VAT: R" << vat << endl;
    cout << "Total to pay: R" << total << endl;
    return 0;
}
```

**[25]**

## QUESTION 3

**3.1** File created and named correctly ✓ (1)

**3.2** #include <iostream> ✓; using namespace std; ✓ (2)

**3.3** int main() ✓; number of students ✓; total and average (double) ✓; highest, lowest and pass counter ✓ (4)

**3.4** Prompt ✓ and cin ✓ in a loop ✓ that repeats while n < 1 or n > 40 ✓ (4)

**3.5** int (or double) array ✓ of size 40 ✓ declared before use ✓ (3)

**3.6** for loop over the students ✓✓; prompt with the student number ✓; cin into marks[i] ✓; inner loop / check ✓ that repeats while the mark is below 0 or above 100 ✓ with an error message ✓; correct index use ✓ (8)

**3.7** Total set to 0 and each mark added ✓✓; average = total / n ✓✓ (with a decimal result) (4)

**3.8** highest and lowest start at marks[0] ✓✓; loop through the array ✓; if marks[i] > highest ✓ then highest = marks[i] ✓; if marks[i] < lowest ✓ then lowest = marks[i] ✓; correct results ✓ (8)

**3.9** Counter set to 0 ✓; if marks[i] >= 50 ✓ the counter increases ✓ inside the loop ✓ (4)

**3.10** Average ✓, highest ✓, lowest ✓ and passes ✓ displayed with labels ✓ (5)

```
// YourName - Question 3
#include <iostream>
using namespace std;

int main() {
    int n = 0, marks[40], highest = 0, lowest = 0, passed = 0;
    double total = 0, average = 0;

    do {
        cout << "How many students wrote (1-40)? ";
        cin >> n;
    } while (n < 1 || n > 40);

    for (int i = 0; i < n; i++) {
        cout << "Mark for student " << i + 1 << ": ";
        cin >> marks[i];
        while (marks[i] < 0 || marks[i] > 100) {
            cout << "Mark must be from 0 to 100." << endl;
            cout << "Mark for student " << i + 1 << ": ";
            cin >> marks[i];
        }
    }

    highest = marks[0];
    lowest = marks[0];
    for (int i = 0; i < n; i++) {
        total = total + marks[i];
        if (marks[i] > highest) highest = marks[i];
        if (marks[i] < lowest) lowest = marks[i];
        if (marks[i] >= 50) passed++;
    }
    average = total / n;

    cout << "Class average: " << average << endl;
    cout << "Highest mark: " << highest << endl;
    cout << "Lowest mark: " << lowest << endl;
    cout << "Students who passed: " << passed << endl;
    return 0;
}
```

**[43]**

**TOTAL: 100**

**END OF MARKING GUIDELINE**